

SQL Complete Practice (Views + LIMIT + Window Functions)

Employee Table (Sample Data)

ID	Name	Dept	Salary
101	Mohan	Admin	4000
102	Rajkumar	HR	3000
103	Akbar	IT	4000
104	Dorvin	Finance	6500
107	Preet	HR	7000
110	Vasudha	IT	7000
111	Melinda	IT	8000
112	Komal	IT	10000
119	Cory	HR	8000
124	Dheeraj	IT	11000

Q1

Create a view for employee name & department and show IT employees

```
CREATE VIEW emp_view AS
SELECT emp_NAME, DEPT_NAME FROM employee;

SELECT * FROM emp_view WHERE DEPT_NAME='IT';
```

Q2

Create view of salary > 5000 and show HR employees

```
CREATE VIEW high_salary AS  
SELECT * FROM employee WHERE salary > 5000;  
  
SELECT * FROM high_salary WHERE dept_name='HR';
```

Q3

Create department average salary view and filter > 6000

```
CREATE VIEW dept_avg AS  
SELECT dept_name, AVG(salary) avg_sal  
FROM employee GROUP BY dept_name;  
  
SELECT * FROM dept_avg WHERE avg_sal > 6000;
```

Q4

Top 5 highest paid employees

```
SELECT * FROM employee ORDER BY salary DESC LIMIT 5;
```

Q5

Employees ranked 6–10 by salary

```
SELECT * FROM employee ORDER BY salary DESC LIMIT 5 OFFSET 5;
```

Q6

Top 10% highest paid employees

```
SELECT * FROM employee
ORDER BY salary DESC
LIMIT (SELECT CEIL(COUNT(*)*0.10) FROM employee);
```

Q7

Assign row number based on salary

```
SELECT emp_name, salary,
ROW_NUMBER() OVER (ORDER BY salary DESC) rn
FROM employee;
```

Q8

Row number within each department

```
SELECT emp_name, dept_name, salary,
ROW_NUMBER() OVER (PARTITION BY dept_name ORDER BY salary DESC) rn
FROM employee;
```

Q9

Rank employees by salary

```
SELECT emp_name, salary,
RANK() OVER (ORDER BY salary DESC) rnk
FROM employee;
```

Q10

Dense rank employees

```
SELECT emp_name, salary,  
DENSE_RANK() OVER (ORDER BY salary DESC) drnk  
FROM employee;
```

Q11

Show department average salary with each row

```
SELECT emp_name, salary,  
AVG(salary) OVER (PARTITION BY dept_name) dept_avg  
FROM employee;
```

Q12

Employees earning more than department average

```
SELECT * FROM (  
  SELECT *, AVG(salary) OVER (PARTITION BY dept_name) avg_sal  
  FROM employee  
) t WHERE salary > avg_sal;
```

Q13

Max salary per department

```
SELECT emp_name, dept_name, salary,  
MAX(salary) OVER (PARTITION BY dept_name) max_sal  
FROM employee;
```

Q14

Employees with max salary in department

```
SELECT * FROM (  
  SELECT *, MAX(salary) OVER (PARTITION BY dept_name) max_sal  
  FROM employee  
) t WHERE salary = max_sal;
```

Q15

Top 2 employees per department

```
SELECT * FROM (  
  SELECT *, ROW_NUMBER() OVER (PARTITION BY dept_name ORDER BY salary DESC) rn  
  FROM employee  
) t WHERE rn <= 2;
```

Q16

Highest paid employee per department

```
SELECT * FROM (  
  SELECT *, ROW_NUMBER() OVER (PARTITION BY dept_name ORDER BY salary DESC) rn  
  FROM employee  
) t WHERE rn = 1;
```

Q17

Top 3 per department AND salary > overall average

```
SELECT * FROM (  
  SELECT *, ROW_NUMBER() OVER (PARTITION BY dept_name ORDER BY salary DESC) rn,  
  AVG(salary) OVER () overall_avg
```

```
FROM employee
) t WHERE rn <=3 AND salary > overall_avg;
```

Q18

Departments with max > 9000 and avg > 5000

```
SELECT dept_name FROM (
  SELECT dept_name,
    MAX(salary) OVER (PARTITION BY dept_name) max_sal,
    AVG(salary) OVER (PARTITION BY dept_name) avg_sal
  FROM employee
) t WHERE max_sal > 9000 AND avg_sal > 5000 GROUP BY dept_name;
```

Q19

Rank + difference from department max

```
SELECT emp_name, salary,
  RANK() OVER (PARTITION BY dept_name ORDER BY salary DESC) rnk,
  MAX(salary) OVER (PARTITION BY dept_name) - salary diff
FROM employee;
```

Q20

GROUP BY vs WINDOW FUNCTION

```
-- GROUP BY
SELECT dept_name, AVG(salary)
FROM employee GROUP BY dept_name;

-- WINDOW FUNCTION
SELECT emp_name, dept_name, salary,
  AVG(salary) OVER (PARTITION BY dept_name) avg_sal
FROM employee;
```

